

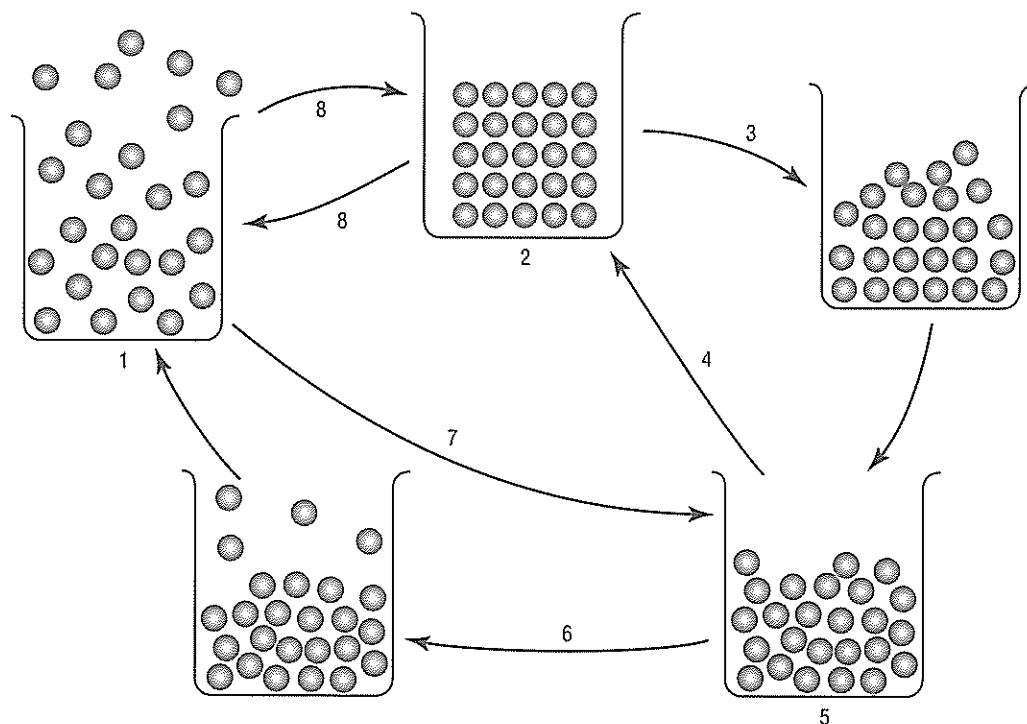
Change of state



Name: _____

Skills: knowledge, understanding

Match the numbers in the diagram to the statements or words below. Write the number in the space.



- a Solid _____
- b Liquid _____
- c Gas _____
- d Melting _____
- e Solidification _____
- f Evaporation _____
- g Condensation _____
- h Sublimation _____
- i Particles being held by strong forces and unable to move away from or past each other _____
- j Particles being held by strong forces but able to move past each other _____
- k Particles escaping from the forces holding them together _____
- l Particles taking up more space as the change of state occurs _____
- m If you heat this it changes into number 5 in the diagram _____
- n If you cool this it becomes number 5 in the diagram _____
- o If you cool this it becomes number 2 in the diagram _____
- p This change explains why clothes dry _____
- q This change explains why dry ice (carbon dioxide solid) does not melt _____

Find the elements



Name: _____

Skills: literacy, knowledge

The names of 22 elements are hidden in this grid. The names go across, down and diagonally and can go backwards or forwards. The elements are from the two lists of elements below the grid.

In the second table the elements are shown only with their symbols. Find the correct spelling of each name, write it in the second table, then find it on the grid. You can use the periodic table from the next activity, Activity 5.5, to find the spelling.

M	U	I	S	E	N	G	A	M	O	O	S
A	Z	A	N	U	B	R	A	C	N	U	O
N	S	D	E	A	L	I	U	T	U	N	D
G	C	A	L	C	I	F	C	F	O	G	I
A	S	E	L	I	C	O	N	R	L	Y	E
N	A	L	U	M	I	N	I	U	M	U	M
E	M	Z	G	K	N	I	Z	U	K	K	S
S	M	U	I	D	O	S	I	L	V	E	R
E	N	O	B	R	A	C	A	V	L	I	S
N	I	T	R	O	L	C	O	P	P	E	R
D	T	U	N	A	S	T	E	N	U	Z	L
E	R	G	C	S	U	L	F	A	N	M	V
N	O	L	H	Y	D	R	O	G	E	E	E
E	G	O	L	X	C	N	X	O	G	R	R
T	E	D	O	A	L	O	Y	L	O	C	S
S	N	E	R	X	E	C	G	D	R	U	U
G	E	N	I	T	K	I	E	Y	D	R	L
N	U	U	N	R	C	L	N	I	Y	Y	V
U	E	O	E	U	I	I	E	G	H	N	E
T	I	I	N	E	N	S	U	L	F	E	R

Aluminium	Carbon	Magnesium	Mercury	Nitrogen	Silicon	Tungsten
Calcium	Hydrogen	Manganese	Nickel	Oxygen	Sulfur	Zinc

Symbol	Name of element
Cl	
Cu	
Au	
Fe	
Pb	
Ag	
Na	
Sn	



Skills: knowledge, interpreting, analysing

	Group I alkaline metals	Group II alkaline earth metals											Group III	Group IV	Group V	Group VI	Group VII	Group VIII	noble gases
Period 1	H hydrogen 1																		He helium 2
Period 2	Li lithium 3	Be beryllium 4											B boron 5	C carbon 6	N nitrogen 7	O oxygen 8	F fluorine 9	Ne neon 10	
Period 3	Na sodium 11	Mg magnesium 12											Al aluminium 13	Si silicon 14	P phosphorus 15	S sulfur 16	Cl chlorine 17	Ar argon 18	
Period 4	K potassium 19	Ca calcium 20	Sc scandium 21	Ti titanium 22	V vanadium 23	Cr chromium 24	Mn manganese 25	Fe iron 26	Co cobalt 27	Ni nickel 28	Cu copper 29	Zn zinc 30	Ga gallium 31	Ge germanium 32	As arsenic 33	Se selenium 34	Br bromine 35	Kr krypton 36	
Period 5	Rb rubidium 37	Sr strontium 38	Y yttrium 39	Zr zirconium 40	Nb niobium 41	Mo molybdenum 42	Tc technetium 43	Ru ruthenium 44	Rh rhodium 45	Pd palladium 46	Ag silver 47	Cd cadmium 48	In indium 49	Sn tin 50	Sb antimony 51	Te tellurium 52	I iodine 53	Xe xenon 54	
Period 6	Cs cesium 55	Ba barium 56	La* lanthanum 57	Hf hafnium 72	Ta tantalum 73	W tungsten 74	Re rhenium 75	Os osmium 76	Ir iridium 77	Pt platinum 78	Au gold 79	Hg mercury 80	Tl thallium 81	Pb lead 82	Bi bismuth 83	Po polonium 84	At astatine 85	Rn radon 86	
Period 7	Fr francium 87	Ra radium 88	Ac** actinium** 89	Rf rutherfordium 104	Db dubnium 105	Sg seaborgium 106	Bh bohrium 107	Hs hassium 108	Mt meitnerium 109	Ds darmstadtium 110	Rg roentgenium 111	Uub ununbium 112							

H

hydrogen

1

chemical symbol

name of element

atomic number

*Lanthanides
58–71**Actinides
90–103

Ce cerium 58	Pr praseodymium 59	Nd neodymium 60	Pm promethium 61	Sm samarium 62	Eu europium 63	Gd gadolinium 64	Tb terbium 65	Dy dysprosium 66	Ho holmium 67	Er erbium 68	Tm thulium 69	Yb ytterbium 70	Lu lutetium 71
Th thorium 90	Pa protactinium 91	U uranium 92	Np neptunium 93	Pu plutonium 94	Am americium 95	Cm curium 96	Bk berkelium 97	Cf californium 98	Es einsteinium 99	Fm fermium 100	Md mendelevium 101	No nobelium 102	Lr lawrencium 103

The periodic table—page 2

Name: _____

The elements are organised in a grid called the periodic table. They are listed in order of their atomic number. They are also arranged so that elements that are alike appear near each other. The elements in a column (called a group) all have similar properties.

Questions

Use the periodic table to answer the following questions.

- 1 State the number of elements in total that are listed in this version of the periodic table. _____
- 2 State the atomic number and symbol for the following elements.

• lead _____ • gold _____ • oxygen _____ • potassium _____ • sulfur _____	• carbon _____ • silver _____ • iron _____ • sodium _____
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- 3 State the name and symbol for atoms with the following atomic numbers:

• 9 _____ • 4 _____ • 40 _____ • 13 _____ • 92 _____	• 80 _____ • 22 _____ • 27 _____ • 15 _____ • 14 _____
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- 4
 - a Record the names of all elements that start with the letter S.

 - b State the symbols for the elements you have chosen.

 - c Explain why some elements have one letter for their symbol and others have two.

- 5
 - a Some columns (or groups) have names given to them. List the names of the three groups that are labelled.

 - b In the group called the noble gases, list the name and symbol of the first three elements in the group.

- 6 Choose any two elements and describe some of their properties.

